ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 4: Consumer Complaint Trends and Response Intelligence Platform

Document Control

Programme	Experience Lab Internship Programme
Week	Week 4
Project	Consumer Complaint Trends and Response Intelligence Platform
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 4 (Thursday, 11:59 PM WAT)

1. PROJECT BACKGROUND

Financial institutions receive complaints from consumers about different financial products and services. These complaints may involve account management, credit cards, mortgages, credit reporting, payment services and other financial activities.

The Consumer Financial Protection Bureau (CFPB) collects consumer complaints and forwards eligible complaints to the relevant companies for review and response.

For this project, AnalystLab Africa is using consumer complaint records relating to Bank of America from 2017 to 2023 to simulate a professional financial services analytics engagement.

The organisation requires a better understanding of complaint trends, frequently reported issues, financial products associated with complaints, recorded company responses and complaint-processing timelines.

Interns will collaborate to develop an integrated complaint intelligence solution using data analysis, statistical methods, predictive modelling, Generative AI and project management practices.

Important: A complaint represents an issue reported by a consumer. It does not automatically establish misconduct or prove that the company has violated any regulation. All findings must be presented objectively and supported by the available data.

2. PROJECT PROBLEM STATEMENT

Stakeholders require a reliable way to monitor consumer complaints and understand patterns in the complaints recorded against Bank of America.

The project must investigate the following business questions:

1. How have complaint volumes changed between 2017 and 2023?
2. Which financial products have the highest complaint volumes?
3. What are the most frequently reported consumer issues?
4. How do reported issues differ across financial product categories?
5. How much time passes between the CFPB receiving a complaint and forwarding it to the company?
6. What company response categories are most frequently recorded?
7. What percentage of complaints has a recorded timely response?
8. Have complaint patterns and company response categories changed over time?
9. Can historical complaint patterns support future complaint-volume forecasting?
10. How can stakeholders access and understand validated complaint insights through an interactive dashboard or AI-powered assistant?

Interns must use evidence from the dataset to answer these questions and document any questions that cannot be answered using the available information.

3. PROJECT OBJECTIVE

Develop a professional, documented and validated consumer complaint intelligence solution that enables stakeholders to:

- Monitor complaint volumes and trends.
- Identify frequently reported financial products and issues.
- Examine company response classifications.
- Analyse CFPB complaint-forwarding intervals.
- Investigate recorded response timeliness.
- Explore historical trends and future complaint-volume forecasting.
- Access validated complaint information through an AI-powered assistant.

The final solution should demonstrate technical competence, analytical reasoning, responsible data use, collaboration and professional communication.

4. DATASET DESCRIPTION

The project uses the following supplied files:

File 1: Consumer_Complaints.xlsx

Contains 62,516 complaint records in one worksheet named `Data`, with 12 columns.

File 2: ConsumerComplaints_DataDictionary.csv

Provides explanations of the available columns and their meanings.

The supplied records cover 1 May 2017 to 28 August 2023.

Available fields

Column	Description
Complaint ID	Unique complaint identifier
Submitted via	Channel used to submit the complaint
Date submitted	Date the CFPB received the complaint
Date received	Date the CFPB sent the complaint to the company
State	Consumer's associated state
Product	Financial product
Sub-product	More specific product classification
Issue	Main reported issue
Sub-issue	Detailed reported issue
Company public response	Recorded public response category
Company response to consumer	Recorded company response or status
Timely response?	Indicator of whether the company provided a timely response

Data preparation requirements

Before conducting analysis, interns must:

1. Inspect the dataset structure and column types.
2. Check the uniqueness of Complaint ID.
3. Identify duplicate records.
4. Investigate missing values.
5. Validate the date columns.
6. Standardise categorical values where necessary.
7. Investigate unusual or inconsistent values.
8. Document data-cleaning decisions.
9. Validate calculations after cleaning.
10. Preserve the original dataset.

Important data interpretation rules:

- Complaint ID is the candidate primary key.
- The supplied file is a flat dataset, not an existing multi-table relational database.
- Date submitted represents when the CFPB received the complaint.
- Date received represents when the CFPB sent the complaint to the company.
- The difference between these dates is the CFPB complaint-forwarding interval, not the company's response or resolution time.
- Missing sub-issue and public response values must be treated according to their meaning and documented.
- Blank Timely responses? values must not automatically be classified as No. In the supplied records, they are associated with complaints marked In progress.

All transformations must be documented and reproducible.

5. TRACK-SPECIFIC ASSIGNMENTS

DATA ANALYTICS

Task: Develop a Consumer Complaint Monitoring and Response Dashboard

Objective: Build an interactive dashboard that enables stakeholders to monitor complaint trends, identify frequently reported issues and examine recorded company responses.

Required Activities

A. Data Inspection and Cleaning

- Import the supplied Excel dataset.
- Inspect the available fields and data types.
- Check duplicate complaint IDs.
- Investigate missing and inconsistent values.
- Validate submission and company-forwarding dates.
- Standardise product, issue and response categories where appropriate.
- Create additional fields for year, month, quarter and forwarding interval.

B. KPI Development

Calculate and document the definitions of:

1. Total complaints.
2. Annual complaint volume.
3. Monthly complaint volume.
4. Complaints by financial product.
5. Complaints by reported issue.
6. Complaints by sub-product and sub-issue.

7. Percentage contribution of each product to total complaints.
8. Company response category distribution.
9. Percentage of complaints marked as receiving a timely response.
10. Number of complaints marked In progress.
11. Average and median complaint-forwarding interval.
12. Year-over-year complaint volume changes.

For timely-response analysis, use only records with a valid Yes or No value in the denominator. Report excluded records separately.

C. Dashboard Development

Build a professional Power BI dashboard containing:

Page 1: Executive Complaint Overview

- Total complaints.
- Complaint trends over time.
- Product distribution.
- Recorded timely-response percentage.
- Average forwarding interval.

Page 2: Product and Issue Analysis

- Complaints by financial product.
- Complaints by sub-product.
- Most frequently reported issues.
- Product-versus-issue analysis.

Page 3: Company Response Analysis

- Company response categories.
- Timely versus untimely response distribution.
- In-progress complaints.
- Public response categories.

Page 4: Complaint Timeline Analysis

- Monthly and annual complaint trends.
- Complaint-forwarding interval distribution.
- Forwarding interval by product.
- Identification of unusually long forwarding intervals.

Include relevant filters, clear titles, KPI definitions and appropriate visualisations.

D. Business Insights

Provide at least five evidence-based findings and three actionable recommendations.

Recommended Tools

Power BI, SQL, Excel or Python.

Required Deliverables

- Cleaned dataset or reproducible cleaning process.
- SQL queries or analytical scripts.
- KPI definitions and calculations.
- Interactive dashboard.
- At least five business insights.
- At least three actionable recommendations.
- README and data-quality documentation.

Assessment Criteria

Criteria	Weight
Business Understanding	10%
Dataset Inspection	15%
Dashboard Design	25%
Data Visualisation	20%
Business Insights & Recommendations	15%
Documentation & Professionalism	15%
Total	**100%**

DATA SCIENCE

Task: Investigate Consumer Complaint Patterns and Response Trends

Objective: Use exploratory data analysis and statistical methods to investigate complaint trends, product categories, reported issues and recorded company responses.

Required Activities

1. Conduct exploratory data analysis.
2. Analyse annual and monthly complaint distributions.
3. Identify frequently reported products and issues.
4. Investigate how complaint patterns have changed over time.
5. Examine the distribution of recorded company responses.
6. Analyse complaint-forwarding intervals.
7. Formulate at least three research questions or hypotheses.
8. Apply appropriate statistical tests where assumptions are satisfied.
9. Interpret the results in the context of financial services.
10. Document limitations and alternative explanations.

Potential research questions include:

- Did complaint volumes differ across the reporting years?
- Are certain products associated with higher complaint volumes?
- Do forwarding intervals vary across financial product categories?
- Are some reported issues more frequently associated with specific company response categories?
- Have recorded response patterns changed over time?

Use suitable techniques such as descriptive statistics, distribution analysis, chi-square tests, non-parametric tests and trend analysis where appropriate.

Do not confuse statistical association with causation.

Recommended Tools

Python, pandas, Jupyter Notebook, NumPy, Matplotlib, Seaborn and SciPy.

Required Deliverables

- Reproducible analysis notebook.
- Data preparation documentation.
- Research questions and hypotheses.
- Exploratory analysis and visualisations.
- Statistical analysis and interpretation.
- At least three supported findings.
- Analytical report and limitations.

Assessment Criteria

Criteria	Weight
Business Understanding & Research Questions	10%
Dataset Inspection & Preparation	15%
Exploratory Data Analysis (EDA) & Visualisation	20%
Statistical Methods & Validation	20%
Findings, Interpretation & Recommendations	15%
Documentation & Reproducibility	10%
Individual Presentation & Communication	10%
Total	**100%**

MACHINE LEARNING

Task: Develop a Consumer Complaint Volume Forecasting Prototype

Objective: Build and evaluate a model that estimates future complaint volumes using historical complaint patterns.

Required Activities

A. Define the prediction problem

Use the historical complaint records to forecast monthly complaint volumes.

Where sufficient records exist, investigate forecasting by selected financial product categories.

B. Prepare the modelling dataset

- Aggregate complaints into monthly counts.
- Investigate missing reporting periods.
- Distinguish missing months from genuine zero-complaint months.
- Create relevant time-based features.
- Investigate historical trends.
- Prepare chronological training, validation and testing periods.
- Prevent information leakage from future periods.

C. Model Development

1. Establish a baseline forecasting approach.
2. Compare it with at least two suitable forecasting methods where feasible.
3. Use chronological testing or rolling-origin validation.
4. Evaluate performance using appropriate metrics such as MAE, RMSE or WAPE.
5. Compare forecast values with actual complaint volumes.
6. Examine prediction errors.
7. Explain the practical implications and limitations.

The first and last reporting months are partial periods and must be handled appropriately.

The forecasting prototype must not claim to predict actual misconduct or prove future company performance.

Recommended Tools

Python, pandas, scikit-learn and appropriate forecasting libraries.

Required Deliverables

- Reproducible modelling notebook or source code.
- Defined prediction target and features.
- Baseline and model comparison.
- Evaluation metrics.
- Forecast visualisations.
- Error analysis.
- Model limitations and recommendations.
- Saved model or reproducible training workflow, where appropriate.

Assessment Criteria

Criteria	Weight
Problem Understanding & Target Definition	10%
Data Preparation & Feature Engineering	15%
Model Development & Comparison	20%
Model Evaluation & Validation	25%
Error Analysis & Business Interpretation	10%
Documentation & Reproducibility	10%
Individual Presentation & Communication	10%
Total	100%

GENERATIVE AI

Task: Build an AI-Powered Consumer Complaint Insights Assistant

Objective: Develop an AI assistant that allows users to ask questions about validated consumer complaint records and receive accurate, data-grounded responses.

Example Questions

- Which financial product recorded the highest number of complaints?
- What were the most frequently reported issues?
- How did complaints change from 2017 to 2023?
- Which company response categories appeared most frequently?
- What percentage of records was marked as receiving a timely response?
- What was the average complaint-forwarding interval?
- Which issues were most frequently associated with a selected product?
- What are the limitations of the supplied complaint records?

Required Activities

1. Review the dataset and data dictionary.
2. Identify questions the dataset can answer.
3. Establish accurate definitions for the key metrics.
4. Build an AI assistant grounded in validated complaint data.
5. Where practical, connect it to controlled SQL queries or analytical functions.
6. Ensure numerical answers come from actual calculations.
7. Make the assistant explain the reporting period and filters used.
8. Test at least 20 questions.
9. Include valid, ambiguous and unsupported questions.
10. Evaluate accuracy, consistency and source grounding.

Important: The supplied dataset does not contain detailed complaint narratives. Therefore, sentiment analysis and narrative-based complaint summarisation are outside the scope of this task.

Recommended Tools

An appropriate LLM platform, Python, a controlled analytics interface or a retrieval-augmented generation approach.

Required Deliverables

- Working AI assistant prototype.
- Documented data sources and KPI definitions.
- System instructions and configuration.
- At least 20 evaluation questions.
- Evaluation results.
- Evidence of grounded numerical responses.
- README, limitations and safeguards.

The assistant must not invent allegations, make legal conclusions or present itself as an official CFPB or Bank of America service.

Assessment Criteria

Criteria	Weight
Business Understanding & Problem Definition	10%
Data Understanding & Knowledge Preparation	15%
AI Solution Design & Implementation	20%
Response Accuracy, Grounding & Evaluation	15%
Responsible AI, Privacy & Limitations	10%
Documentation & Reproducibility	10%
Individual Presentation & Demonstration	10%
Total	**100%**

PROJECT MANAGEMENT

Task: Plan and Coordinate Delivery of the Consumer Complaint Intelligence Platform

Objective: Plan, coordinate and monitor the delivery of the integrated project across all five internship tracks.

Required Activities

A. Project Initiation

- Develop a project charter.
- Define objectives, scope and expected outcomes.
- Identify stakeholders.
- Establish deliverables and acceptance criteria.
- Document assumptions and dependencies.

B. Project Planning

- Develop a Work Breakdown Structure.
- Create a project schedule with task owners and milestones.
- Identify cross-track dependencies.
- Prepare a RACI matrix.
- Develop a stakeholder communication plan.

C. Risk and Issue Management

Create a risk register addressing:

- Incomplete or inconsistent complaint records.
- Incorrect date calculations.
- Misinterpretation of company responses.
- Incorrect complaint counts.
- Forecasting limitations.
- Inaccurate AI-generated responses.
- Delays in cross-track deliverables.
- Incomplete integration and documentation.

Include probability, impact, mitigation, owner and status for each risk.

D. Project Monitoring

- Maintain a task and milestone tracker.
- Record progress, blockers and decisions.
- Coordinate peer reviews.
- Monitor delivery against the project schedule.
- Track acceptance criteria.
- Coordinate integration and final review.
- Prepare progress reports.

E. Final Delivery
Coordinate the handover of the dashboard, statistical findings, forecasting prototype, AI assistant and supporting documentation where those outputs have actually been completed and tested.

Recommended Tools

Excel, Microsoft Project, Trello, Notion or another suitable project management tool.

Required Deliverables

- Project charter.
- Work Breakdown Structure.
- Gantt chart or project schedule.
- RACI matrix.
- Risk and issue register.
- Stakeholder communication plan.

- Task and milestone tracker.
- Final project status report.
- Evidence of coordination and delivery.

Assessment Criteria

Criteria	Weight
Project Understanding & Charter Development	15%
Work Breakdown Structure & Project Scheduling	20%
Stakeholder Management & Role Allocation	10%
Risk, Issue & Change Management	15%
Project Monitoring, Coordination & Execution	20%
Project Reporting & Final Handover	10%
Documentation & Individual Presentation	10%
Total	100%

6. SHARED INTEGRATION AND COLLABORATION REQUIREMENTS

All five tracks contribute to one overall project, but every intern remains responsible for their assigned track's technical deliverables.

The team must establish consistent definitions for complaint volume, product, issue, company response, timely-response indicator and complaint-forwarding interval.

The final project should bring together the following outputs where technically appropriate:

1. **Complaint Monitoring Dashboard:** Validated KPIs, trends and product comparisons.
2. **Statistical Analysis:** Evidence-based findings and interpretations.
3. **Predictive Prototype:** Evaluated complaint-volume forecasts.
4. **Complaint Insights Assistant:** Natural-language access to validated information.
5. **Project Documentation:** Scope, milestones, risks, testing, integration evidence and final handover.

Interns must show evidence of actual integration and explain how their work supports other tracks.

7. FINAL SUBMISSION REQUIREMENTS

Each intern must submit:

1. Track-specific deliverables and working files.
2. A README explaining the project, methodology, tools, setup instructions, assumptions and limitations.
3. Evidence of data cleaning, validation and testing.
4. Findings, recommendations or technical solutions relevant to the assigned track.
5. Evidence of individual contribution and collaboration.
6. A record of meaningful revisions made after feedback.

7. An individual recorded presentation of approximately 5–10 minutes explaining the work, results, limitations and contribution to the overall project.

All files must be properly organised and named so mentors can inspect the work and reproduce relevant results where applicable.

8. ASSESSMENT AND QUALITY EXPECTATIONS

Interns will be assessed according to their assigned track.

Assessment will consider:

- Understanding of the business problem.
- Technical execution and methodology.
- Data quality, testing and validation.
- Accuracy and usefulness of the solution.
- Interpretation of findings and recommendations.
- Integration and collaboration.
- Documentation and reproducibility.
- Individual presentation.

Technical requirements must be assessed according to each discipline rather than applying unrelated criteria across all tracks.

9. DEFINITION OF SUCCESS

The project will be considered successful when interns deliver technically sound, documented and validated solutions that demonstrate how consumer complaint records can support better complaint monitoring and business understanding.

The completed solution should provide reliable complaint reporting, identify recurring products and issues, examine recorded company responses, investigate complaint-forwarding patterns and demonstrate appropriate predictive or AI-enabled capabilities.

Every conclusion must be supported by evidence. Interns must distinguish consumer-reported issues from verified facts and must not interpret recorded company responses as proof of satisfactory complaint resolution.

The overall project should demonstrate professional problem-solving, responsible data use, reproducible work, effective communication and multidisciplinary collaboration.

PROJECT SOURCE AND ATTRIBUTION

Dataset: Consumer Complaints on Financial Products and Services — Bank of America (2017–2023).

Data Provider: Consumer Financial Protection Bureau (CFPB).

Supplied Files:

- Consumer_Complaints.xlsx
- ConsumerComplaints_DataDictionary.csv

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 4 project.
- Publish a LinkedIn post describing your project.
- Explain how your project schedule supports successful project execution.
- Publish an X (Twitter) post summarizing your project.
- Tag **AnalystLab Africa** on LinkedIn.
- Tag **@analystlabafric** on X.
- Include the hashtag **#AnalystLabAfrica** on both platforms.

Professional Branding Reminder

Successful projects depend on effective planning and scheduling. Project Managers must be able to organize work, allocate resources efficiently, identify critical activities, and communicate realistic timelines to stakeholders. This assignment is designed to simulate real-world project scheduling responsibilities and help you build practical skills that are highly valued across industries.